

PhytoGraph — cycles 11-13

2026-05-18

Contents

PhytoGraph — cycles 11-13	1
Abstract	1
Introduction	1
Approach	2
Findings	2
Discussion	6
Open Questions	7
References	7
Appendix: Implementation Details	8

PhytoGraph — cycles 11-13

Abstract

Cycles 11-13 repaired campaign bookkeeping, completed two missing Wave 3 instruments, rebuilt the Botanical Atlas as a six-track integration surface, and then reconciled a post-merge disagreement so Track 1 and Track 3 were no longer stale Atlas placeholders.

The main technical result is that the Atlas now exposes all six track-local Wave 3 outputs across 60,000 searchable taxon pages with evidence-vs-prediction labels. Track 1 contributes 60,000 tree-compatibility index rows. Track 3 contributes pending convergence-prior rows for drupe and capsule. Tracks 2, 4, 5, and 6 remain exposed through their existing adapters. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` remain header-only.

This cycle range does not close Barrier 3 at audit level and does not validate any biological prediction. It makes the instruments queryable from the Atlas and preserves the campaign boundary that track-local hypotheses are not master-ledger claims.

Introduction

PhytoGraph is organized around barriers. Barrier 2 had already produced a frozen substrate and track enrichment. Wave 3 then required predictive instruments and an Atlas surface that could query those instruments without blurring observed evidence, enriched evidence, model predictions, and missing data.

At the end of cycle 10, three problems remained. Track 1 had an implemented tree-compatibility instrument but lacked the required report. Track 3 had score tables and diagnostics but lacked the required report. The Atlas scaffold existed as a contract, but the generated site outputs and Track 1/Track 3 queryability were not yet demonstrated. Cycles 11-13 addressed those gaps.

The reporting sources for this cycle range are:

- Cycle 11 worker session `0b9e8647-ca47-48dc-9089-532600e91768`.

- Cycle 12 researcher session 60693f48-7d7a-43e8-9ddc-d9048f53f167.
- Cycle 13 worker session 0b3756f0-b117-4880-8478-8f96258bfd2a.
- Fork eec13528227c merge reports.
- The post-merge integration audit report provided with this task.
- Workspace artifacts listed in the appendix.

Approach

The work proceeded in three steps.

First, cycle 11 repaired plan and ledger alignment. A manager intervention had identified hard `promise_check` failures caused by Wave 4 child milestone IDs that existed in the append-only ledger but not in `plan_of_record.md`. The worker added exact plan rows for M4.V2, M4.V5, and M4.A-track5-duke-source-ablation, then reran `promise_check` and `org_check`. This was a book-keeping repair only. It did not advance Barrier 3 and did not promote predictions.

Second, cycle 12 launched a three-branch Wave 3 fan-out under fork eec13528227c:

Branch	Objective	Required artifact
Clone 0	Build Track 1 Reticulation Atlas instrument and report	tracks/track1/reports/track1_reticulation_atlas.md
Clone 1	Build Track 3 Convergence Pressure instrument and report	tracks/track3/reports/track3_convergence_pressure.md
Clone 2	Build Botanical Atlas scaffold	botanical_atlas_site/reports/atlas_barrier3_scaffold.md

Third, cycle 13 integrated the fork outputs into the main Atlas surface. The post-merge worker changed the Atlas builder and tests, regenerated the site outputs, and confirmed that Track 1 and Track 3 were no longer represented as stale placeholder contracts.

Findings

Cycle 11: Plan-Ledger Repair

Cycle 11 repaired a consistency failure, not a biological or instrument gap. The problem was that three Wave 4 ledger rows already existed under milestone IDs that were not declared in the plan of record. This caused `promise_check` to fail.

The worker updated `plan_of_record.md` with the exact child milestone IDs:

- M4.V2
- M4.V5
- M4.A-track5-duke-source-ablation

The repair note states that these are historical track-local outputs and do not authorize additional Wave 4 validation before Barrier 3. After the update, `promise_check` exited 0, and both master ledgers remained header-only.

Cycle 12: Track 1 Reticulation Atlas

Track 1 now has the required Wave 3 Reticulation Atlas report and regenerated TCI outputs. TCI means tree-compatibility index: a per-taxon score in $[0,1]$ where lower values indicate

more evidence that a single-parent tree encoding loses information.

The Track 1 worker repaired the evidence boundary in `tracks/track1/instruments/build_tci.py`. Observed reticulation evidence now counts only accepted-key, event-shaped evidence: hybridization_event, polyploidization_event, event-shaped reticulate_inheritance_evidence, or accepted multi-parent crop_pedigree evidence. Chromosome-count and ploidy-state rows are structural context only; they no longer count as direct reticulation events.

The instrument produced:

Output	Count
Accepted keys scored	60,000
Evidence-supported taxa	2
Structural-only taxa	29,294
Data-limited unknown taxa	30,704
Genus hotspot rows	14,292
Accepted-key resolved Track 1 enrichment rows	3
Pending-crosswalk Track 1 enrichment rows	25

The TCI summary reports a minimum TCI of 0.5, mean TCI of 0.986858, and maximum TCI of 1.0. The two evidence-supported taxa are driven by retained multi-parent crop-pedigree evidence. The original canonical polyploid seed cases remain data-limited because they are pending-crosswalk or absent in the frozen accepted-key namespace. *Arachis hypogaea* and *Avena sativa* recover through accepted multi-parent crop-pedigree evidence, but the report explicitly says this does not validate the original M1.3 canonical seed set.

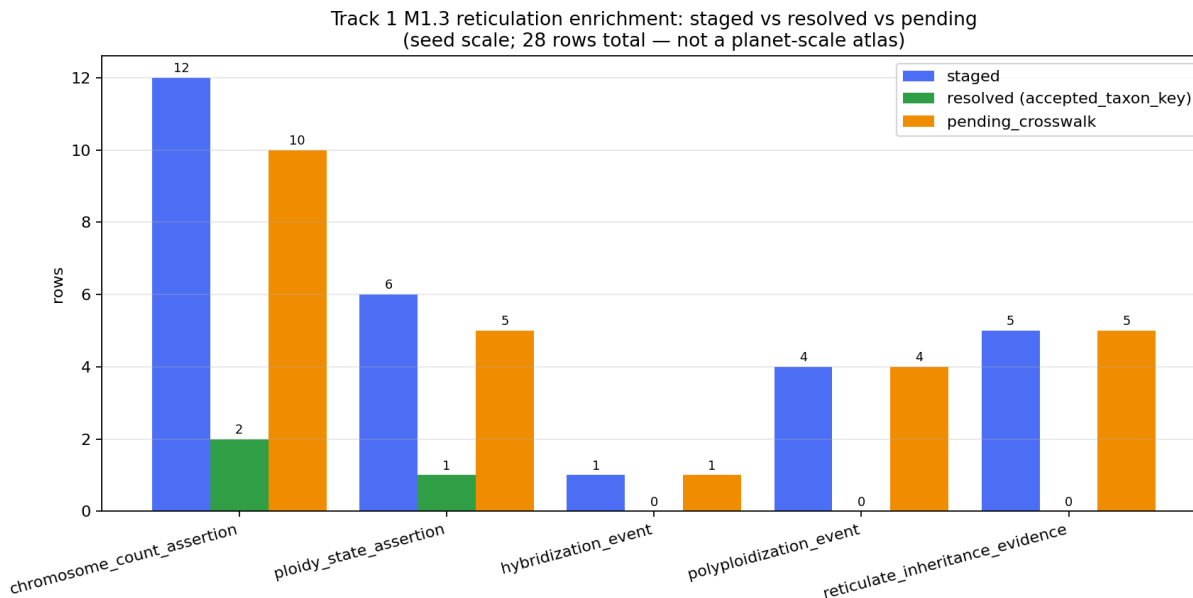


Figure 1: Track 1 reticulation coverage by edge type. The key point is that direct event evidence remains sparse; structural context dominates current coverage.

Track 1 validation run reported 28 passing focused reticulation tests, Barrier 2 conformance PASS, and `promise_check/org_check` exit 0 with known warnings only.

Cycle 12: Track 3 Convergence Pressure

Track 3 now has the required Wave 3 Convergence Pressure report and a track-local prediction TSV for Atlas ingestion. Convergence pressure is a trait-level statistic that asks whether a retained trait is more dispersed across families than expected under two null models. The report uses $CP_{min}(T) = \min(z_{N1}(T), z_{N2}(T))$, where N1 preserves family-size carrier loads and N2 preserves sampling density through per-taxon Track 3 edge-count weights.

A trait clears the pending convergence-pressure bar when $CP_{min} \geq 2.0$. In the current frozen substrate, two traits clear that bar:

Trait	Accepted-key carriers	Families	CP_min	Status
drupe	187	32	5.647	pending convergent-trait hypothesis
capsule	544	48	4.831	pending convergent-trait hypothesis

The `convergence_predictions.tsv` row classes are:

Row class	Count
pending_convergent_trait_hypothesis	2
observed_trait_evidence_summary	10
data_limited_canonical_trait	3
diagnostic_bucket_excluded	1

The `_other` bucket remains diagnostic-only. The data-limited canonical traits are `ant_domatia`, `carnivory`, and `parasitism`, each with zero accepted-key carriers in the frozen Track 3 enrichment.

The falsifier summary records strong confounding in raw entropy but not a configured failure of the standardized statistic:

Measure	Value
R2_observed_H_family	0.835
R2_CP_min	0.852
spearman_rho_residOBS_vs_CPmin	0.406
Falsifier verdict	PASS

The report interprets this narrowly: Track 3 does not validate H3, but it also does not falsify H3 under the stated $abs(\rho) > 0.8$ threshold. Wave 4 must still validate the two pending hypotheses against independent trait lists and source/family/sampling ablations. Track 3 uses AusTraits-heavy retained trait coverage from the earlier enrichment layer [31].

Track 3 validation run reported 16 passing focused tests, Barrier 2 conformance PASS, Barrier 1 substrate validation PASS, and header-only master ledgers.

Cycle 12: Botanical Atlas Scaffold

The Atlas branch built a static site scaffold under `botanical_atlas_site/`. Its purpose is to make track outputs visible and queryable, not to generate new predictions.

The scaffold produced:

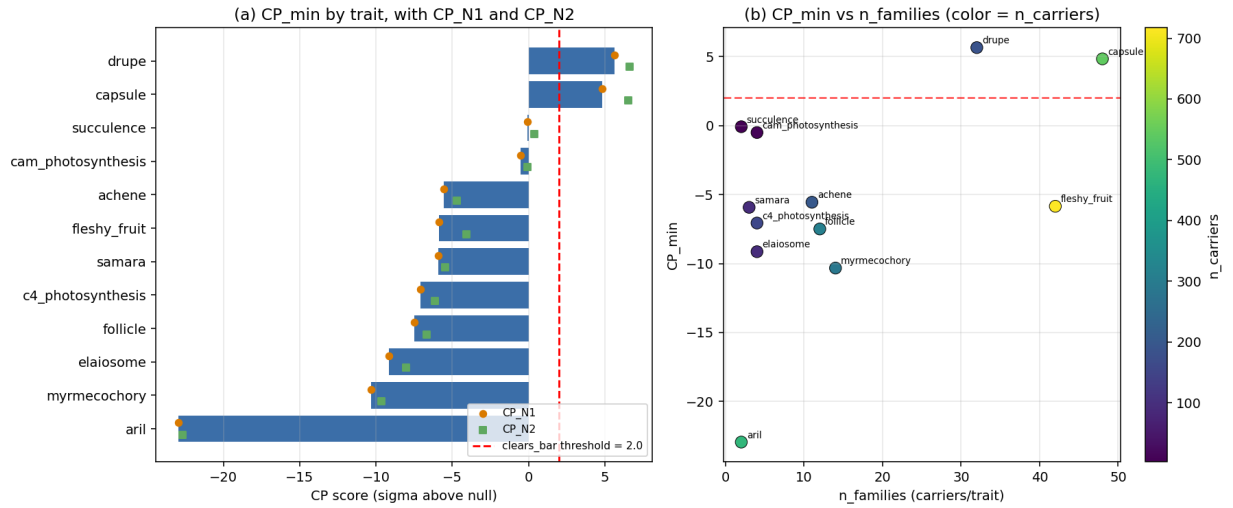


Figure 2: Track 3 convergence-pressure figure. The plot shows the two traits clearing the current pending CP threshold and the diagnostic relationship to family-count structure.

Artifact	Role
index.html, app.js, style.css	Local static Atlas shell
search_index.json	60,000-row search index
pages/*.json	60,000 per-taxon page payloads
coverage_summary.json	Per-track coverage and adapter status
provenance_registry.json	Source-group provenance summary
counter_claim_template.json	Counter-claim schema
build_log.json	Reproducible build metadata
tools/file_counter_claim.py	Targeted counter-claim validator
tests/test_atlas_build.py	Atlas contract tests

The scaffold enforced four page states:

State	Meaning
observed	Source-ingested substrate rows
enriched	Wave 2 track-local source projections
predicted	Wave 3 instrument or result rows
data-limited	No accepted-key substrate or instrument row for that taxon/track section

In the initial clone-2 branch, Track 2, Track 4, Track 5, and Track 6 outputs were exposed, while Track 1 and Track 3 were contract placeholders. The branch therefore did not close Barrier 3 by itself.

Cycle 13: Post-Merge Atlas Integration

Cycle 13 reconciled the post-merge disagreement: the Atlas no longer leaves Track 1 and Track 3 as stale placeholders. The worker integrated Track 1 TCI outputs and Track 3

convergence-prior outputs into the Atlas adapters, rebuilt the site, and reran the checks.

The final Atlas build reported:

Metric	Value
Substrate nodes read	363,237
Substrate hyperedges read	641,183
Taxon crosswalk rows read	75,269
Per-taxon pages written	60,000
Search-index rows written	60,000
Build elapsed time	61.43 seconds

Final per-track page state:

Track	Observed pages	Enriched pages	Predicted pages	Data-limited pages
Track 1	0	0	60,000	0
Track 2	0	0	6	59,994
Track 3	4	3,198	50	56,748
Track 4	0	0	5	59,995
Track 5	0	1,255	65	58,680
Track 6	0	0	48	59,952

Final Atlas instrument rows detected by adapter:

Track	Instrument rows	Taxa
Track 1	60,000	60,000
Track 2	6	6
Track 3	50	50
Track 4	6	5
Track 5	1,405	65
Track 6	244	48

The post-merge tests reported 20 passing tests across Atlas, Track 1, and Track 3. Barrier 2 conformance passed. Barrier 1 substrate validation passed with 363,237 nodes and 641,183 retained hyperedges. `promise_check` and `org_check` exited 0 with warnings only.

One nonblocking warning remains: `promise_check` warns that ledger line 146 tracks `botanical_atlas_site/pages/` as a directory artifact rather than a canonicalized file artifact. The check still exits 0.

Discussion

Cycles 11-13 changed the campaign state from “Wave 3 outputs exist but are not fully surfaced” to “all six track-local outputs are queryable from the Atlas.” This matters because Barrier 3 requires instrument-to-Atlas integration, not just standalone instrument files.

The work remains conservative in three ways.

First, the master ledgers are still empty. `prediction_ledger.tsv` and `speculation_ledger.tsv` each contain only a header row. Track-local rows are visible in the Atlas, but they have not been reconciled into Barrier 4 master prediction statuses.

Second, the Atlas labels Track 1 and Track 3 rows as predictions or instrument outputs with caveats. Track 1 TCI rows are data-limited instrument results, not established reticulation claims. Track 3 drupe and capsule rows are pending convergence-prior hypotheses, not validated convergence findings.

Third, Barrier 3 is ready for assessment but not closed by the worker reports. The post-merge integration made the missing outputs queryable, but the audit report explicitly says no audit-level validation was performed.

Open Questions

The immediate open question is whether an auditor should close Barrier 3 after inspecting the updated Atlas. The worker evidence supports queryability and evidence-vs-prediction labeling, but audit-level closure remains unperformed.

Track 1 still needs accepted-key repair for the original canonical polyploid/hybrid seeds before recovery rates can be interpreted. Current recovery is data-limited for the original seed set.

Track 3 still needs Wave 4 validation and ablations for drupe and capsule. The statistic generated pending hypotheses, not validated convergence biology.

The Atlas still has a known limitation in Track 3 page projection: it uses explicitly listed supporting hyperedges from `convergence_predictions.tsv`, and long support lists are truncated there. The pages are queryable, but not exhaustive carrier enumerations.

Track 4 and Track 6 remain data-limited in their own ways. Track 4 lacks observed bioclim vectors. Track 6 executed deterministic offline controls only and did not evaluate hosted or local foundation models.

References

- [4] The World Flora Online Consortium et al., “World Flora Online Plant List December 2025,” Zenodo, 2025. <https://doi.org/10.5281/zenodo.18007552> (accessed 2026-05-17).
- [27] CCDB, “Chromosome Counts Database (CCDB),” Tel Aviv University, version 1.66.6, 2026. <https://ccdb.tau.ac.il/> (accessed 2026-05-17).
- [28] Anna Rice, Lior Glick, Shiran Abadi, et al., “The Chromosome Counts Database (CCDB) — a community resource of plant chromosome numbers,” *New Phytologist*, 2015. <https://doi.org/10.1111/nph.13191> (accessed 2026-05-17).
- [29] Ilia J. Leitch, Emma Johnston, Jaume Pellicer, Oriane Hidalgo, and Michael D. Bennett, “Plant DNA C-values Database,” Royal Botanic Gardens, Kew, 2026. <https://cvalues.science.kew.org/> (accessed 2026-05-17).
- [30] Troy E. Wood, Naoki Takebayashi, Michael S. Barker, et al., “The frequency of polyploid speciation in vascular plants,” *Proceedings of the National Academy of Sciences*, 2009. <https://doi.org/10.1073/pnas.0811575106> (accessed 2026-05-17).
- [31] Daniel Falster, Rachael Gallagher, Elizabeth Wenk, Herve Sauquet, et al., “AusTraits: a curated plant trait database for the Australian flora,” Zenodo, version 6.0.0, 2024. <https://doi.org/10.5281/zenodo.11188867> (accessed 2026-05-17).

Appendix: Implementation Details

Source Inventory

Source ID	Date	Contents	Timeline role
0b9e8647-ca47-48dc-9089-532600e91768	2026-05-18T03:31:33Z	Cycle 11 worker summary for plan-ledger namespace repair	Repaired promise_check failures before further Wave 3 work
60693f48-7d7a-43e8-9ddc-d9048f53f167	2026-05-18T03:41:41Z	Cycle 12 researcher brief	Directed three-branch fan-out for Track 1, Track 3, and Atlas
.long-exposure/fork-eec13528227c/clone-0/merge_report.md	2026-05-18	Track 1 clone merge report	Documents M3.T1 instrument, outputs, and tests
.long-exposure/fork-eec13528227c/clone-1/merge_report.md	2026-05-18	Track 3 clone merge report	Documents M3.T3 statistic, outputs, and tests
.long-exposure/fork-eec13528227c/clone-2/merge_report.md	2026-05-18	Atlas clone merge report	Documents M3.A scaffold before post-merge Track 1/3 integration
.long-exposure/fork-eec13528227c/fanout_merge.md	2026-05-18	Three-branch merge summary	Confirms all required branch deliverables existed
0b3756f0-b117-4880-8478-8f96258bfd2a	2026-05-18T04:09:07Z	Cycle 13 worker summary	Documents post-merge Atlas integration and checks
Provided audit report	2026-05-18	Post-merge integration audit summary	Confirms tests and validators after integration

Code Organization

Area	Key files	Purpose
Plan/ledger repair	plan_of_record.md, promise_ledger.jsonl	Aligns milestone IDs and append-only events
Track 1	tracks/track1/instruments/build_tci.py, tracks/track1/reports/track1_reticulation_atlas.md, tracks/track1/outputs/*.tsv	Builds and reports TCI instrument

Area	Key files	Purpose
Track 3	tracks/track3/scripts/convergence_pressure.py, tracks/track3/reports/track3_convergence_pressure.md, tracks/track3/data/convergence_*	Builds and reports convergence-pressure statistic
Atlas	botanical_atlas_site/build_atlas.py, index.html, app.js, style.css, coverage_summary.json, pages/*.json	Static site generation and per-track adapters
Counter-claims	tools/file_counter_claim.py, counter_claim_template.json	Targeted correction payload validation
Tests	tests/test_atlas_build.py, tracks/track1/tests/test_tci_instrument.py, tracks/track3/tests/test_convergence_pressure.py	Contract and instrument checks

Test Results

Check	Result
Cycle 11 promise_check after plan repair	exit 0
Cycle 11 org_check	exit 0 with known warnings
Track 1 focused reticulation pytest set	28 passed
Track 3 focused tests	16 passed
Atlas scaffold tests in clone 2	10 passed
Post-merge Atlas + Track 1 + Track 3 tests	20 passed
Barrier 2 conformance after post-merge integration	PASS
Barrier 1 substrate validation after post-merge integration	PASS: 363,237 nodes; 641,183 retained hyperedges
Post-merge promise_check	exit 0; warnings only
Post-merge org_check	exit 0; known root-layout warnings only

File Counts And Rows

Artifact	Count
tracks/track1/outputs/tci_per_taxon.tsv	60,000 data rows
tracks/track1/outputs/tci_hotspots_genus.tsv	14,292 data rows
tracks/track1/outputs/canonical_recovery_report.tsv	12 data rows
tracks/track3/data/convergence_predictions.tsv	16 data rows
tracks/track3/data/convergence_pressure_scores.tsv	16 data rows
tracks/track4/data/crop_substitution_candidates.tsv	3 data rows
tracks/track6/data/probe_results.tsv	840 data rows
botanical_atlas_site/pages/*.json	60,000 files

Artifact	Count
prediction_ledger.tsv	0 data rows
speculation_ledger.tsv	0 data rows

Cross-Reference Map

Origin	Consuming artifact	Flow
tracks/track1/instruments/build_tci.py	tracks/track1/outputs/tci_per_taxon.tsv	Computes TCI rows from frozen substrate and Track 1 enrichment
tracks/track1/outputs/tci_per_taxon.tsv	botanical_atlas_site/pages/*.json	Exposed by the Atlas as Track 1 predicted/instrument rows
tracks/track3/scripts/convergence_pressure.py	tracks/track3/data/convergence_predictions.tsv	Emits Track 3 row classes for pending hypotheses and evidence summaries
tracks/track3/data/convergence_predictions.tsv	botanical_atlas_site/pages/*.json	Exposed by the Atlas as Track 3 predicted/enriched/observed/data-limited states
tracks/track2, tracks/track4, tracks/track5, tracks/track6 track-local outputs	botanical_atlas_site/coverage_summary.json	Preserved by adapters during post-merge Track 1/3 integration
tests/test_atlas_build.py	botanical_atlas_site/reports/atlas_barrier3_scaffold.md	Tests the six fixed track sections, provenance, inferred flags, search rows, and counter-claim validation
MANIFEST.md	This report	Updated as the concise cycle 11-13 workspace snapshot while preserving the existing ## Key Files section